

Bahrrouni Ayoub

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SUMMARY

AI Engineering student and Full-Stack Developer with a strong background in Computer Systems Engineering, IoT, Embedded Systems, and AI-powered applications. Skilled in designing scalable end-to-end systems across web, mobile, cloud, and embedded environments. Experienced with AI/ML, full-stack development, IoT architecture, backend systems, and real-time data-driven applications using Python, React, Node.js, Flutter, MongoDB, PostgreSQL, ESP32, MQTT, and LangChain technologies.

EXPERIENCE

IoT & Software Development Intern

Tunisie Télécom

March 2025 - June 2025, Ariana, Tunisia

- **Engineered** a smart irrigation system integrating IoT sensors, ESP32 microcontrollers, and AI models for predictive water management.
- **Developed** a cross-platform **Flutter mobile app** and a **Node.js + MongoDB backend** for real-time data visualization and analytics.
- **Integrated** MQTT protocols to ensure reliable, low-latency device-to-cloud communication and optimized system responsiveness.
- **Enhanced** operational efficiency by leveraging predictive analytics to reduce water waste and improve irrigation scheduling.
- **Documented** system architecture and deployment procedures, ensuring scalability and future maintainability.
- **Key Skills Demonstrated**
 - IoT Development (**ESP32, MQTT, sensors**)
 - Mobile Development (**Flutter**)
 - Backend Development (**Node.js, MongoDB**)
 - Frontend Development (**React.js**)
 - Cloud Integration and Optimization (**scalable storage, real-time analytics**)

PROJECT

PFE – AI-Driven Smart Irrigation

Tunisie Télécom • github.com/BahrrouniAyoub/PlantCare_Flutter.git • March 2025 - June 2025

- Designed and implemented a **smart irrigation system** integrating IoT sensors, ESP32, and MQTT for real-time monitoring.
- Built a **Flutter-based mobile app** and **Node.js + MongoDB backend** to visualize data and optimize water usage.
- Integrated basic AI models to predict irrigation needs, improving efficiency.
- **Skills Used:** ESP32, MQTT, Flutter, Node.js, MongoDB, React.js, AI basics

CodeQuest – Byte Blases Academy

Github: <https://github.com/BahrrouniAyoub/byte-blades-academy.git>

- Building a retro RPG inspired coding education platform with quests XP progression guilds and interactive learning experiences.
- Tech Stack: React, TypeScript, Node.js, PostgreSQL, Supabase, Phaser.js, Tailwind CSS

PagePal

Github: <https://github.com/BahrrouniAyoub/pagepal.git>

- Built an AI powered voice PDF reader that converts books into natural voice narration.
- Implemented authentication PDF processing user libraries and AI voice services.
- Tech Stack: Next.js, TypeScript, MongoDB.js, Clerk, Vapi AI, Tailwind CSS

AyoubOS – VSCode Inspired Developer Portfolio

Github: <https://github.com/BahrrouniAyoub/vscode-portfolio.git>

- Built an immersive VSCode style developer portfolio with terminal emulation command platte and AI Copilot
- Integrated OpenRouter APIs and portfolio aware AI assistant workflows.

EDUCATION

AI Engineering - ESPRIT (present)

Bachelor's Degree in Computer Systems Engineering – Specializing in IoT and Embedded Systems

Minor in Full-Stack development • Faculty of Sciences of Tunis • Rue de Tolède, Tunis 2092 20 • 2025

CERTIFICATIONS

- Full-Stack Development Certification — 3W Academy
- JavaScript Certification – freeCodeCamp
- Responsive Web Design Certification – freeCodeCamp
- LLM Engineering in Practice with Streamlit and OpenAI – 365 DataScience
- Data Cleaning and Preprocessing with pandas – 365 DataScience

COURSEWORK

Networking and Communication Systems

Faculty of Sciences of Tunis • 2025

- Understanding TCP/IP stack and wireless communication. Designing and testing network topologies. Basic network security principles.
- Configured IoT devices for stable network communication during testing phases.
- Simulated a **Smart House network** using Cisco Packet Tracer to model device interactions.

Mobile App Development (Flutter, Android Studio)

Faculty of Sciences of Tunis • 2025

- Cross-platform mobile UI design and API integration. State management and user authentication. Testing and deploying mobile applications.
- Built the **Flutter-based mobile app** for your PFE, integrating real-time IoT data visualization.
- Developed an **Android health app** with secure authentication and a responsive interface.

IoT Architecture and Protocols (MQTT, HTTP, TCP/IP)

Faculty of Sciences of Tunis • 2024

- Designing communication layers for IoT systems. Using lightweight protocols for efficient data transfer. Securing and optimizing device-to-cloud communication
- Integrated **MQTT protocol** in your Smart Irrigation PFE to ensure reliable real-time data exchange.
- Used **HTTP APIs** for data transmission between devices and cloud services in other prototypes.

Microcontroller Programming (ESP32, STM32, Arduino)

Faculty of Sciences of Tunis • 2024

- Configuring microcontrollers and peripherals. Sensor calibration and integration. Power management and hardware troubleshooting.
- Programmed **ESP32 boards** to monitor soil humidity and temperature in your PFE.
- Built sensor-driven logic for your **Smart Garage Door** and **home automation simulations**.

Database Design and Management (SQL, NoSQL)

2023 • Database schema design and normalization. CRUD operations and query optimization. Using NoSQL for unstructured IoT data.

- Implemented **MongoDB backend** for your Smart Irrigation and health app projects.
- Designed relational databases for a **Library Management System** in Java.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, PHP, R, Embedded C, HTML, CSS

Frontend: React, Next.js, Vite, Tailwind CSS, Redux, React Router, Framer Motion

Backend: Node.js, Express.js, FastAPI, Flask, Django, REST APIs, JWT, WebSocket

AI/ML: LangChain, LangGraph, RAG Pipelines, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy

Databases: PostgreSQL, MongoDB, Redis, Neo4j, SQLite, MySQL, Firebase, Supabase

IoT & Embedded: ESP32, STM32, Arduino, Raspberry Pi, MQTT, TCP/IP

Tools: Docker, Kubernetes, Git, GitHub Actions, Linux, AW

